

关于欧洲空域更改无线电通信失效程序的运行提示

背景

在对公司欧洲航路紧急程序的梳理中发现，欧洲航空安全局（EASA）在 2025 年 5 月开始执行的新的欧洲标准航空程序（SERA）中，对无线电通信失效（RCF）程序做出了新的规定和要求。

由于我司 747 机型在欧洲空域执行的北线/南线航路均涉及相关程序，飞行部已按照流程对欧洲标准航空程序（SERA）的相关规定和要求进行了梳理，并据此发布运行提示。

ESRA 程序要求

SERA 8035

a. 作为受控飞行操作的飞机，应在适当的通信频率上保持持续的空地语音通信监听，并在必要时与相应的 ATC 建立双向通信，但经相关 ANSP（空中导航服务供应商）规定的在受控机场作为机场交通一部分的飞机除外。

1.当已经建立 CPDLC 时，飞机维持空地语音通信监听的要求仍然有效。

SERA 14083

a. 当飞机无法遵守 SERA.8035 条款（a）时，飞行机组应尝试在之前使用的频道建立联系，如不成功，则在当前航路的其它合适的频道尝试。如果仍然失败，飞行机组应尝试以下对象建立通信：

- 1.合适的空中交通服务单位；
- 2.其它空中交通服务单位；或
- 3.其它飞机，

飞行机组应尝试通过所有可用手段建立通信，包括但不限于数据链、卫星电话和移动电话，并在成功建立联系后，告知其无法在指定频道建立联系的状况。

b. 当在预定时间内未收到来自某架飞机的通信，从而怀疑可能发生通信故障，或收到其他空中交通服务单位的请求时，空中交通管制员应在飞机可能监听的频率上呼叫该飞机，并且：

1. 在提供监视服务时，空中交通管制员通常应确定飞机的接收机是否正常工作，如果成功，应继续通过 SSR 代码/ADS-B 数据变化或 IDENT 发送来提供空中交通管制服务，以获取对发送给飞机的管制许可的确认；

2. 如果不成功，空中交通管制单位应：

i. 请求其他空中交通服务单位通过呼叫该飞机提供帮助，必要时转报；

ii. 请求航路上的飞机尝试与该飞机建立通信，必要时转报；

iii. 尽快向航空器运营人通报任何空地通信故障；

3. 如果上述第 2 条第 i 款和 ii 款中描述的尝试失败，除非经发送人特别要求，否则不得向该飞机盲发空中交通管制许可；其他信息应在飞机可能正在接收的频率上通过盲发方式发送。

c. 当飞行前无法遵守 SERA.8035 条款(a)，并且上述 SERA.14083 条款(a)描述的建立通讯的尝试未成功时，使用下列无线电通信失效程序：

1. 当在管制机场飞行时，应注意观察可能通过目视信号发布的指令。

2. 将应答机设置为 A7600 和/或将 ADS-B 发射机设置为显示无线电空地通信失效的代码，并按情况遵守第 3、4、5 和 6 条中描述的程序。

3. VFR 飞行器应保持在 VMC 飞行，在最近合适机场着陆，并通过最快捷的方式向相关空中交通服务单位报告其到达。

4. 除第 5 条另有规定外，IFR 飞行器应当：

i. 在下列情况发生后 20 分钟内，保持最后指令的速度和高度，或最低飞行高度，取较高者：

A. 飞机未能按要求发送报告；或

B. 在监视服务提供的情况下，应答机设置为 7600 和/或发送适当的 ADS-B 危急和/或紧急代码后，

随后根据已提交的飞行计划，并结合延误和修正的信息，调整高度和速度；

ii. 当被雷达引导，或在 RNAV 航路中按 ATC 指令执行偏置飞行时：

A. 若有限制指令，按该限制继续飞行，然后重新加入最近接收到并确认的航路，同时考虑适用的最低飞行高度；或

B.若无限制指令，应在不迟于下一重要点前重新加入最近接收到并确认的航路，同时考虑适用的最低飞行高度。

iii.按照最近收到并确认的航路许可，直飞服务于目的地机场的合适指定导航台或定位点，并按需按照下文第 iv 条的要求，在该导航台或定位点盘旋，直到开始下降；

iv.从第 iii 条指定的导航台或定位点开始下降，时间是最近收到并确认的预计进近时间（或尽可能接近该时间）；如果未收到并确认预计进近时间，则按 ETA（或尽可能接近 ETA 的时间）开始下降。

v.完成和指定导航台或定位点匹配的仪表进近程序；且

vi.如果可能，应在上述第 iv 条提及的 ETA 或最后确认的预期进近时间（以较晚者为准）后的 30 分钟内置陆。

5.在 SID 或 STAR 中飞行的飞机，应按照相应的航图上所规定的无线电通信失效程序执行（如提供）。

6.如果执行 IFR 的飞机遇到 VMC 且机长决定继续保持 VMC 飞行，机长应调定应答机编码 A7601，在最近合适机场着陆，并通过最快捷的方式向相关空中交通服务单位报告其到达。

d. 为在相关空域内的飞行器提供空中交通管制服务的前提是，飞机在发生通信故障时遵守条款（c）的规定

e. 一旦得知在其责任区内飞行的某架飞机出现明显的无线电通信失效，空中交通管制单位应将关于通信失效的信息传达给沿航线的所有相关空中交通服务单位。目的地机场所在区域的区域管制中心应采取措施，获取备用机场及飞行计划中列出的其他相关信息，如果此类信息不可用。

f. 当一个空中交通管制单位收到信息，得知一架飞机在通信失效后已重新建立通信或已着陆，该单位应通知飞机发生通信故障时所在区域的空中交通管制单位，以及飞行航路上的其他相关空中交通服务单位，提供继续管制所需的必要信息，以便飞机继续飞行。

g. 通信失效情况下使用的信号应符合附录 1 的相关规定。

操作建议

1. 飞行机组应知晓相关紧急程序，并明确紧急程序的查询方法、尽快查阅理解相关法规；此外，对 SERA 应急程序更新的内容、以及与其它地区相关程序（或 ICAO 标准程序）的差异应当有明确的认识。

2. 飞行机组应当意识到, 747 机队文件中对相关程序的总结不是标准化操作程序; 总结性质的文件仅作为辅助性资料, 不具备法规效力, 在紧急情况下需要使用怎样的处置程序, 完全基于飞行机组的决断, 也是飞行机组的基本责任之一。

3. 相关程序查询路径: JEPP FD-Europe Airway Manual-Emergency-Emergency Procedures-Europe-RCF Procedures Europe

附件: SERA RCF 程序原文

Radio Communication Failure Procedures Europe

Ref Regional Supplementary Procedures EASA SERA.8035 and SERA.14083

EASY ACCESS RULES FOR STANDARDISED EUROPEAN RULES OF THE AIR (SERA)

Radio Communication Failure Procedures

SERA.8035	a. An aircraft operated as a controlled flight shall maintain continuous air-ground voice communication watch on the appropriate communication channel of, and establish two-way communication as necessary with, the appropriate air traffic control unit, except as may be prescribed by the relevant ANSP in respect of aircraft forming part of aerodrome traffic at a controlled aerodrome. 1. The requirement for an aircraft to maintain an air-ground voice communication watch shall remain in effect when CPDLC has been established.
SERA.14083	a. When an aircraft is unable to comply with SERA.8035, point a., the flight crew shall attempt to establish contact on the previous channel used and, if not successful, on another channel appropriate to the route. If these attempts fail, the flight crew shall attempt to establish communication with: 1. the appropriate air traffic services unit; 2. other air traffic services units; or 3. other aircraft, using all available means, including, inter alia, data link, satellite voice and mobile phones and, when successful, advise that contact on the assigned channel could not be established. b. When an expected communication from an aircraft has not been received within a time period such that the occurrence of a communication failure is suspected, or when requested by other air traffic services units, the air traffic controller shall call the aircraft on the frequencies on which the aircraft is believed to be listening, and: 1. when providing surveillance service, the air traffic controller shall normally determine whether or not the aircraft's receiver is functioning, and if successful, continue providing air traffic control service using SSR code/ADS-B transmission changes or IDENT transmissions to obtain acknowledgement of clearances issued to the aircraft; 2. if not successful, the air traffic control unit shall: (i) request other air traffic services units to render assistance by calling the aircraft and relaying messages, if necessary; (ii) request aircraft on the route to attempt to establish communication with the aircraft and relay messages, if necessary; (iii) initiate the notification to the aircraft operator, as soon as possible, of any failure in air-ground communication; 3. if the attempts described in points 2.(i) and 2.(ii) fail, blind transmission of air traffic control clearances shall not be made to aircraft, except at the specific request of the originator. Other messages should be transmitted by blind transmission on the frequencies on which the aircraft is believed to be listening.

- c. When an aircraft is unable to comply with point SERA.8035(a) and the attempts described in point (a) of SERA.14083 to establish communications are not successful, the radio communication failure procedures described below shall be applied:
 1. The aircraft, when forming part of the aerodrome traffic at a controlled aerodrome, shall keep a watch for instructions as may be issued by visual signals.
 2. The aircraft shall set the transponder on Mode A Code 7600 and/or set the ADS-B transmitter to indicate the loss of air-ground communications and comply with the procedures described in points 3., 4., 5. and 6., as appropriate.
 3. A VFR flight shall continue to fly in visual meteorological conditions, land at the nearest suitable aerodrome, and report its arrival by the most expeditious means to the appropriate air traffic services unit.
 4. Except as provided for in point 5., an IFR flight shall:
 - (i) maintain the last assigned speed and level, or minimum flight altitude if higher, for a period of 20 minutes following:
 - A. the aircraft's failure to make a required report; or
 - B. the time the transponder is set to 7600 and/or the appropriate ADS-B emergency and/or urgency mode is transmitted if surveillance service is provided,

and thereafter adjust level and speed in accordance with the filed flight plan as amended by delay and modification messages to the filed flight plan;
 - (ii) when being vectored or having been directed by ATC to proceed offset using area navigation (RNAV):
 - A. with a specified limit, continue to that limit, then rejoin the last received and acknowledged route, taking into consideration the applicable minimum flight altitude; or
 - B. without a specified limit, rejoin the last received and acknowledged route no later than the next significant point, taking into consideration the applicable minimum flight altitude;
 - (iii) proceed according to the last received and acknowledged route clearance to the appropriate designated navigation aid or fix serving the destination aerodrome and, when required to ensure compliance with point (iv) below, hold over this aid or fix until commencement of descent;
 - (iv) commence descent from the navigation aid or fix specified in point (iii) at, or as close as possible to, the expected approach time last received and acknowledged; or, if no expected approach time has been received and acknowledged, at, or as close as possible to, the estimated time of arrival;
 - (v) complete an instrument approach procedure as specified for the designated navigation aid or fix; and
 - (vi) land, if possible, within 30 minutes after the estimated time of arrival specified in point (iv) or the last acknowledged expected approach time, whichever is later.
 5. An IFR flight following a standard instrument departure route or a standard instrument arrival route shall comply with the procedures for radio communication failure specified on the Standard Departure Chart – Instrument (SID) or Standard Arrival Chart – Instrument (STAR), when provided.
 6. If an IFR flight encounters visual meteorological conditions and the pilot-in-command decides to continue to fly in visual meteorological conditions, the pilot shall set Mode A Code 7601, land at the nearest suitable aerodrome, and report arrival by the most expeditious means to the appropriate air traffic services unit.
- d. The provision of air traffic control service to flights operating in the airspace concerned shall be based on the premise that an aircraft experiencing communication failure complies with point c.
- e. As soon as it is known that an aircraft operating in its area of responsibility is experiencing an apparent radio communication failure, an air traffic control unit shall forward information concerning the radio communication failure to all air traffic services units concerned along the route of flight. The area control centre in whose area the destination aerodrome is located shall take steps to obtain information on the alternate aerodrome(s) and other relevant information specified in the filed flight plan, if such information is not available.

	f. When an air traffic control unit receives information that an aircraft, after experiencing a communication failure, has re-established communication or has landed, that unit shall inform the air traffic control unit in whose area the aircraft was operating at the time the failure occurred, and other air traffic services units concerned along the route of flight, giving necessary information for the continuation of control if the aircraft continues its flight. g. The signals used in case of communication failure shall be in accordance with Appendix 1.
GM1 SERA.14083(b) (1)	ATC UNIT ATTEMPT TO ESTABLISH COMMUNICATION WITH AIRCRAFT Some aircraft equipped with first-generation ADS-B avionics do not have the capability of squawking IDENT while the emergency and/or urgency mode is selected.
GM1 SERA.14083(c)	RADIO COMMUNICATION FAILURE PROCEDURE — MULTIPLE AIRCRAFT The air-ground voice communication failure procedures specifically address failure of communications affecting a single aircraft. In situations where multiple aircraft may be involved, especially when operating in an environment where high-frequency communications are used as primary means for voice communication, the safety of operations is best assured when aircraft adhere to the last ATC clearance received and acknowledged, similarly to the case of ground radio station failure.
GM2 SERA.14083(c)	LANDING AT THE NEAREST SUITABLE AERODROME When assessing the suitability of an aerodrome for landing, the pilot should consider runway characteristics, aerodrome facilities and the complexity of the operating environment at that aerodrome.
GM1 SERA.14083(c) (4)(i)(B)	AIRCRAFT ADS-B CAPABILITIES Some aircraft equipped with first-generation ADS-B avionics can transmit a general emergency alert only, regardless of the code selected by the pilot.